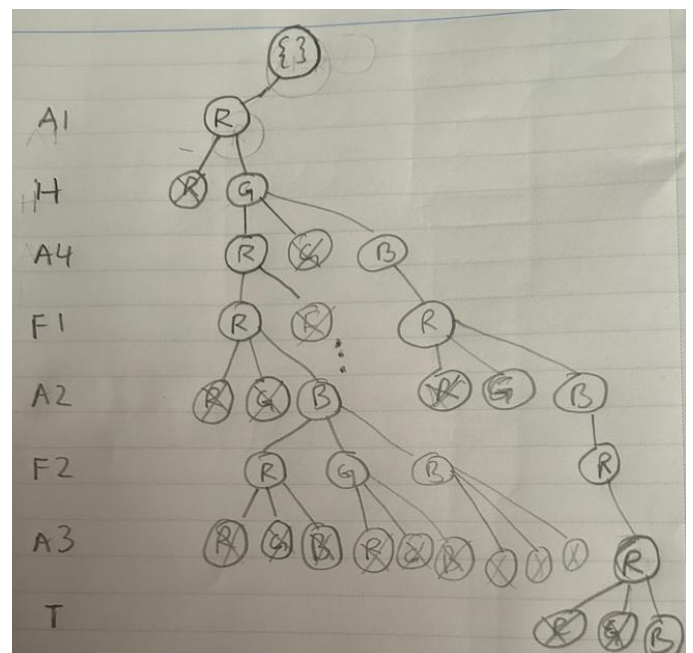
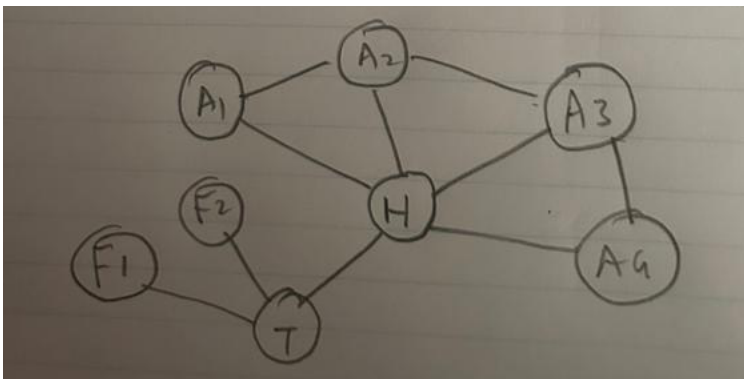


CSP

Question 1

Consider the graph with 8 nodes **A1, A2, A3, A4, H, T, F1, F2**. **A_i** is connected to **A_{i+1}** for all **i**, each **A_i** is connected to **H**, **H** is connected to **T**, and **T** is connected to each **F_i**. Find a 3-coloring of this graph by hand using the following strategy:
Backtracking with variable order **A1, H, A4, F1, A2, F2, A3, T**, and the value order **R, G, B**.

Solution



Question 2

Explain why it is a good heuristic to choose the variable that is most constrained but the value that is least constraining in a CSP search?

Solution

The most constrained variable makes sense because it chooses a variable that is (all other things being equal) likely to cause a failure, and it is more efficient to fail as early as possible (thereby pruning large parts of the search space). The least constraining value heuristic makes sense because it allows the most chances for future assignments to avoid conflict.

Question 3

You have four cards that you want to send to your friends A,B,C,D,E,F,G,H,I,J,K,L,M and N, with this restriction that no two friends who know each other get the same card.

A is friend with E and J and H.
B is friend with E
C is friend with J and N
D is friend with M
F is friend with D and M and K
G is friend with N
I is friend with D and M
L is friend with D
H is friend with J

Solution

```
>>> {'A': 1, 'C': 1, 'B': 1, 'E': 2, 'D': 1, 'G': 1, 'F': 2, 'I': 2, 'H': 3, 'K': 3, 'J': 4, 'M': 4, 'L': 2, 'N': 2}
```

Question 4

Crossword puzzle as Constraint satisfaction problem

1		2		3
	4			
		5		

Variables : X_1, X_2, X_3, X_4, X_5

The position of the variables in the puzzle is labeled 1,2,3,4,5

Domain values:

X_1 can be assigned {hoses, laser, sheet, snail, steer}

X_2 can be assigned {hike, aron, keet, earn, same}

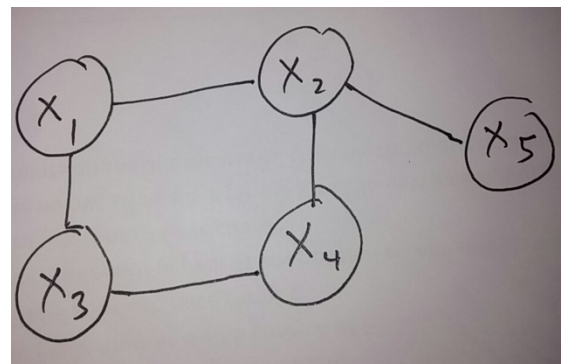
X_3 can be assigned {run, sun, let}

X_4 can be assigned {hike, aron, keet}

X_5 can be assigned {no, be}

PART A

Draw constraint graph



PART B

Solve the problem using constraint satisfaction using MRV heuristic, degree heuristic and forward checking.

Solution

X5 = no <- start with X5 since it is the most constraint variable (MRV heuristic)

X2 = aron <- forward checking lets you know that X4 has no legal values left so
backtrack

X2 = earn

X4 = aron <- one legal value left (MRV)

X1 = sheet < backtrack as x3 has no legal value

X1 = steer

X3 = run <- one legal value left

Question 5

Four TAs (**Mark, Mike, Rob, Sam**) are trying to write eight questions for a final exam (somewhat like this one):

1. **C**onstraints
2. **O**ptimal Search
3. **G**ames
4. **R**ules
5. ID-Trees
6. **N**eural Nets
7. **S**VMs
8. **B**oosting

Some questions are harder to write than others, so they should be distributed about equally. And some problems are so similar to others that for variety, we don't want those written by the same TA. This leads to some constraints:

- $TA(\text{Constraints}) \neq TA(\text{Rules})$
- $TA(\text{Boosting}) \neq TA(\text{ID-Trees})$
- $TA(\text{Boosting}) \neq TA(\text{SVMs})$
- $TA(\text{SVMs}) \neq TA(\text{Optimal Search})$
- $TA(\text{SVMs}) \neq TA(\text{Games})$
- $TA(\text{SVMs}) \neq TA(\text{Rules})$
- $TA(\text{SVMs}) \neq TA(\text{Neural Nets})$
- $TA(\text{Neural Nets}) \neq TA(\text{Games})$

Also, there are some demands that have to be honored.

- Mike insists on doing Boosting
- Mike will not do Constraints
- Mark insists on doing Games
- Only Rob willing to do SVMs.

Part A

Draw lines between variables with a $\neq$ constraint, and use the TAs' demands to reduce domains by crossing out names. Continue to reduce domains using the constraints while possible

Part B

Solve the problem using constraint satisfaction using MRV heuristic, degree heuristic and forward checking.

Solution

SVM – Rob <- MRV heuristic + degree heuristic

Boosting – Mike <- MRV heuristic

Game – Mark <- MRV heuristic

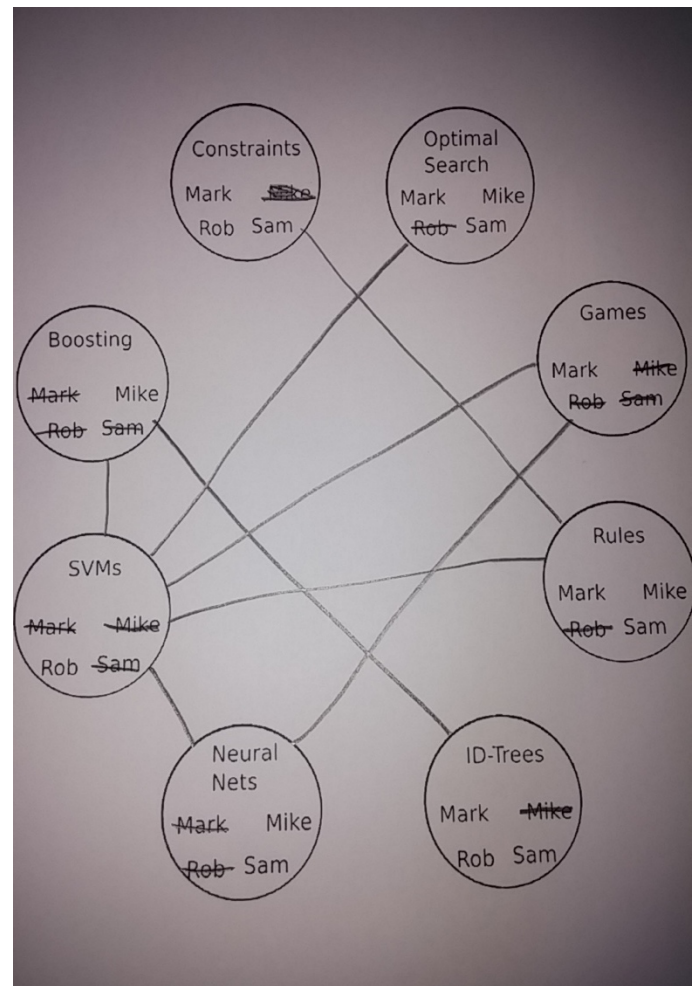
Neural Network - Mike

Rules – Sam

Constraint – Rob

ID-trees – Sam

Optimal search - Mark



Question 6.

You are tasked to make a jingle of 8 measures. Each measure can have one chord value from {A, B, C, D, E, F}. You can think of it as constraint satisfaction problem with each measure a variable and possible chord values as domain.

Part a

There are constraints that establish **initial domain values**:

1. The jingle must start with chord A.
2. The jingle must end with chord A or F
3. Measure 7 must have chord E
4. Chord A cannot appear at any position except at first and last measure.
5. Measure 4 must have chord E.

In the table below, show the possible initial domain values for each measure, as determined by the constraints on specific measures. Use only the constraints that establish initial domain values.

Measure	Possible Initial Domain Values
1	A
2	B C D E F
3	B C D E F
4	E
5	B C D E F
6	B C D E F
7	E
8	A , F

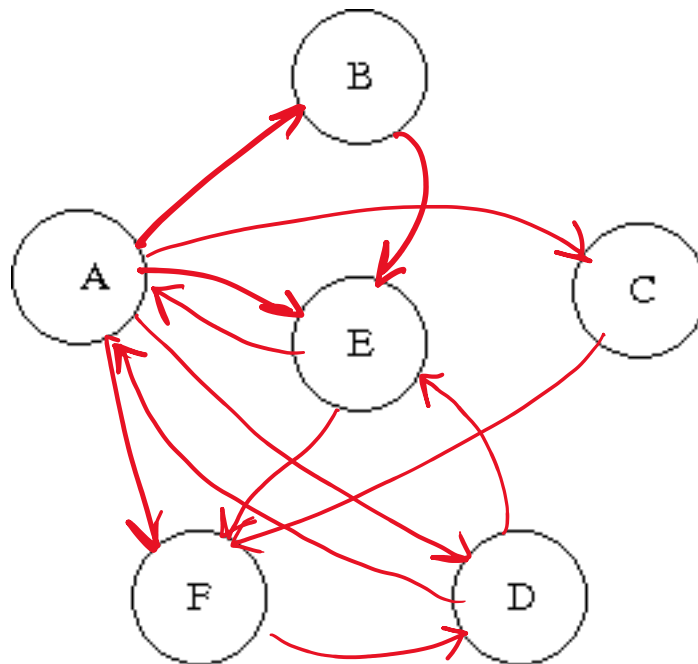
Part B

There are also constraints on chord transitions; that is, these are the **constraints between variables**.

1. $A \rightarrow \text{any other chord}$
2. $B \rightarrow E$
3. $C \rightarrow F$
4. $D \rightarrow A \text{ or } E$
5. $E \rightarrow A \text{ or } F$
6. $F \rightarrow D$
7. Two consecutive measures cannot have the same chords

For example, the first constraint read as: If chord A is assignment to a measure, any chord can be assigned to the next measure.

Fill out the following graph with directed arrows showing the allowed chord transitions. Note that this graph is merely another way of representing the constraints between adjacent measures.

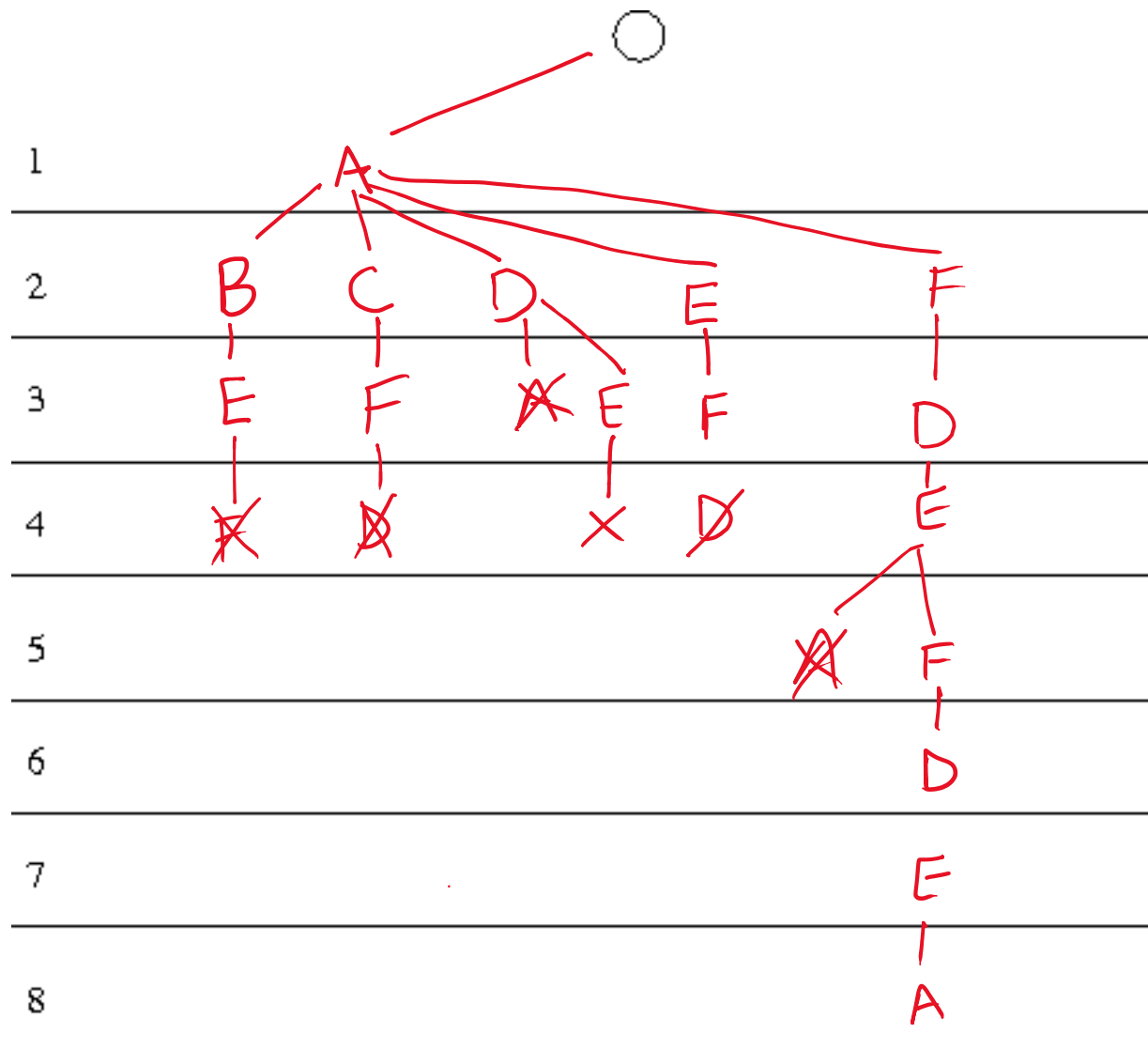


This is a different way of writing the constraints using domain values as nodes. In exam use variables as node for constraint graph

Part C

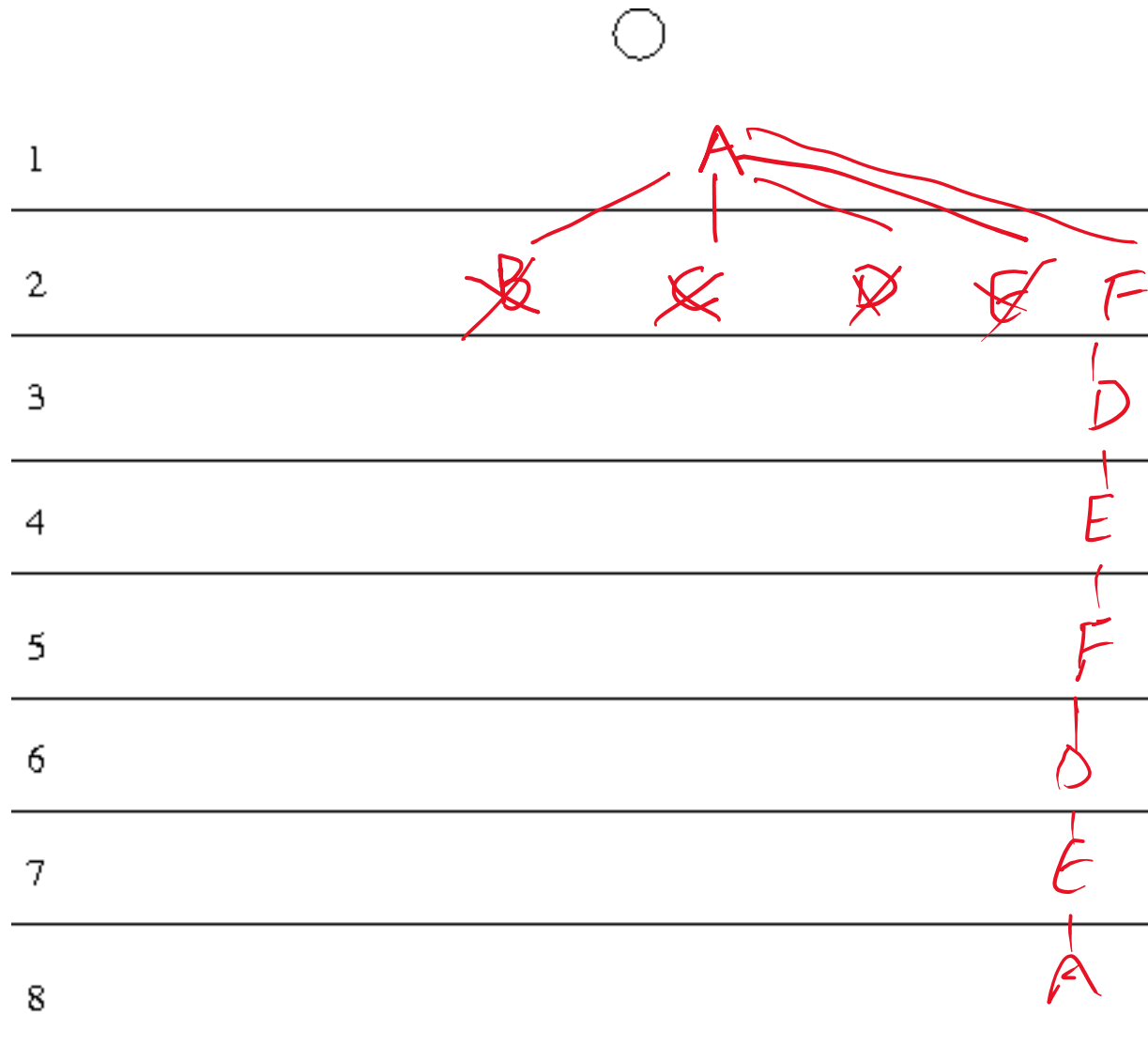
Fill in the tree for a backtracking search that finds a set of assignments of chords to measures that satisfies all the constraints.

- Use constraints of Part A and Part B
- Search proceeds, as shown by numbered levels, from measure 1 to measure 8.
- You try chords in lexical order, from A to F
- As usual, any path whose last node violates any constraint is terminated.



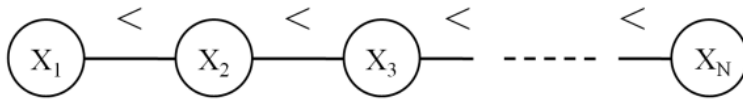
Part D

Repeat Part C using **Forwards chaining**, i.e. keep track of remaining legal values for unassigned variables and terminate a path which may lead to constraint violation



Question 6

Consider the general less-than chain CSP below. Each of the N variables X_i has the domain $\{1 \dots M\}$. The constraints between adjacent variables X_i and X_{i+1} require that $X_i < X_{i+1}$



For now, assume $N = M = 5$.

1. [1 pt] How many solutions does the CSP have?

Only one solution

2. [1 pt] What will the domain of X_1 be after enforcing the consistency of only the arc $X_1 \rightarrow X_2$?

$\text{Dom}(X_1) = \{1, 2, 3, 4\}$

3. [1 pt] What will the domain of X_1 be after enforcing the consistency of only the arcs $X_2 \rightarrow X_3$ then $X_1 \rightarrow X_2$?

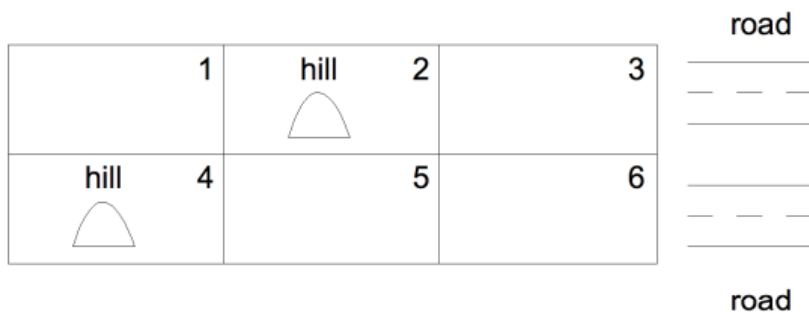
$\text{Dom}(X_1) = \{1, 2, 3\}$

Question 7

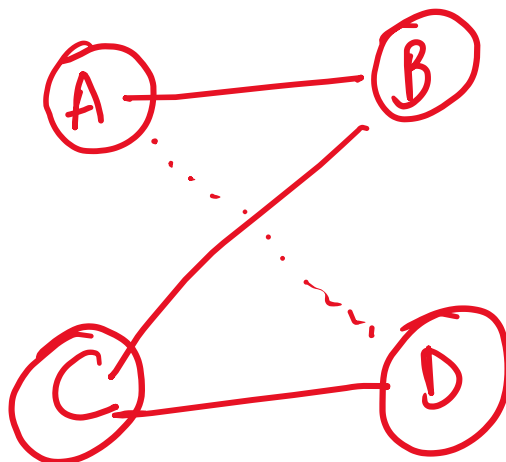
You are asked to determine the layout of a new, small college. The campus will have four structures: an administration building (A), a bus stop (B), a classroom (C), and a dormitory (D). Each building must be placed somewhere on the grid below. The following constraints must be satisfied:

- (i) The bus stop (B) must be adjacent to the road.
- (ii) The administration building (A) and the classroom (C) must both be adjacent to the bus stop (B).
- (iii) The classroom (C) must be adjacent to the dormitory (D).
- (iv) The administration building (A) must not be adjacent to the dormitory (D).
- (v) The administration building (A) must not be on a hill.
- (vi) The dormitory (D) must be on a hill or near the road.
- (vii) All buildings must be in different grid squares.

Here, “adjacent” means that the buildings must share a grid edge, not just a corner.



1. [1 pts] Express the non-unary above constraints using constraint graph. Use solid line to indicate adjacency constraint and dotted line to indicate non-adjacency constraint.



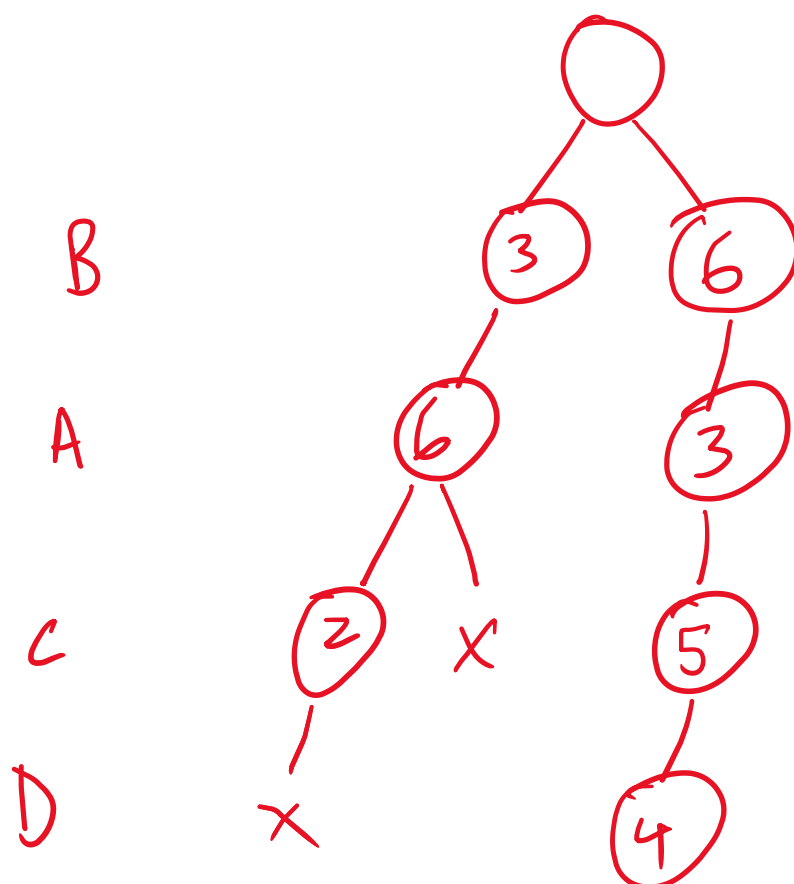
2. [4 pts] Cross out eliminated values to show the domains of all variables after unary constraints and arc consistency has been applied in the following order:

$$A \rightarrow B, B \rightarrow A, A \rightarrow D, D \rightarrow A, B \rightarrow C, C \rightarrow B, C \rightarrow D, D \rightarrow C$$

A	[	1	2	3	4	5	6	]
B	[	1	2	3	4	5	6	]
C	[	1	2	3	4	5	6	]
D	[	1	2	3	4	5	6	]

— using unary constraint
 // using arc consistency

3. [2pts] Give a solution for this CSP or state that none exist. **Draw the search tree.** Use **Minimum remaining heuristic** and **degree heuristic**. Break the ties using alphabetical order.



Solution:

- A - 3
- B - 6
- C - 5
- D - 4

Question

Consider the following CSP with 3 variables **X**, **Y** and **Z**: **Dom(X)** = {1, ..., 10}, **Dom(Y)** = {5, ..., 15}, and **Dom(Z)** = {5, ..., 20}, and binary constraints: **C(X,Y)** : $X > Y$, **C(Y,Z)** : $Y + Z = 12$, and **C(X,Z)** : $X + Z = 16$.

a) [1 pts] Is the arc $X \rightarrow Y$ consistent? Why/ Why not?

No, for values 1,2,3,4,5 there is not domain value for Y such that $C(X,Y) : X > Y$ is satisfied

b) [1 pts] Is the arc $X \rightarrow Z$ consistent? Why/ Why not?

Yes, because for all values of X there is some domain value for Z to satisfy $C(X,Z) : X + Z = 16$